

Exercise 2: Implementing the Factory Method Pattern

Code:

```
public class FactoryMethodPatternExample{
    public static void main(String[] args){
        Document d1 = DocumentFactory.createDocument("word");
        d1.open();
        Document d2 = DocumentFactory.createDocument("pdf");
        d2.open();
        Document d3 = DocumentFactory.createDocument("excel");
        d3.open();

    }
}

interface Document{
    void open();
}

class DocumentFactory{
    public static Document createDocument(String s){
        if(s.equalsIgnoreCase("word")){
            return new WordDocument();
        }
        else if(s.equalsIgnoreCase("pdf")){
            return new PdfDocument();
        }
        else if(s.equalsIgnoreCase("excel")){
            return new ExcelDocument();
        }
        else{
            return new WordDocument();
        }
    }
}

class WordDocument implements Document{
    public WordDocument createDocument(){
        return new WordDocument();
    }
    public void open(){
```

```
        System.out.println("Word Document Opened!");
    }
}

class PdfDocument implements Document{
    public PdfDocument createDocument(){
        return new PdfDocument();
    }
    public void open(){
        System.out.println("PDF Document Opened!");
    }
}

class ExcelDocument implements Document{
    public ExcelDocument createDocument(){
        return new ExcelDocument();
    }
    public void open(){
        System.out.println("Excel Document Opened!");
    }
}
```

Output:

Module-1 > Design Patterns and Principles > FactoryMethodPatternExample.java > WordDocument > open()

```
17 class DocumentFactory{
18     public static Document createDocument(String s){
19         return new ExcelDocument();
20     }
21 }
22
23 else{
24     return new WordDocument();
25 }
26
27 }
28
29 }
30
31 }
32
33
34 class WordDocument implements Document{
35     public WordDocument createDocument(){
36         return new WordDocument();
37     }
38     public void open(){
39         System.out.println(x: "Word Document Opened!");
40     }
41 }
42
43 class PdfDocument implements Document{
44     public PdfDocument createDocument(){
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\adits\Desktop\DeepSkilling\Coding> ^C
PS C:\Users\adits\Desktop\DeepSkilling\Coding>
PS C:\Users\adits\Desktop\DeepSkilling\Coding> c:: cd 'c:\Users\adits\Desktop\DeepSkilling\Coding'; & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\adits\AppData\Roaming\Code\User\workspaceStorage\3385de95c309e825d680cb68353935b\redhat.java\jdt_ws\Coding_613b24a0\bin' 'FactoryMethodPatternExample'
Word Document Opened!
PDF Document Opened!
Excel Document Opened!
PS C:\Users\adits\Desktop\DeepSkilling\Coding>

powerShell
powerShell
Run: Factory...